15 Discovery Way, Acton, MA 01720
Phone: (978)263-3584, **Fax:** (978)263-5086
Web Site: www.princetoninstruments.com

Acton SpectraHub™

Detector Interface Operating Instructions

Rev. 06.08.09
SpectraHub manual.doc

Table of Contents

I. Product Description

The Acton SpectraHub Detector Interface is a microprocessor-based single channel 20-bit data acquisition module for use with SpectraSense software. All Acton detectors and preamps with DC current or voltage output as well as photon counting detectors are compatible with the SpectraHub. Through SpectraSense, which is supplied with the unit, the SpectraHub allows scanning, plotting and saving spectral data with any Acton SP series monochromator.

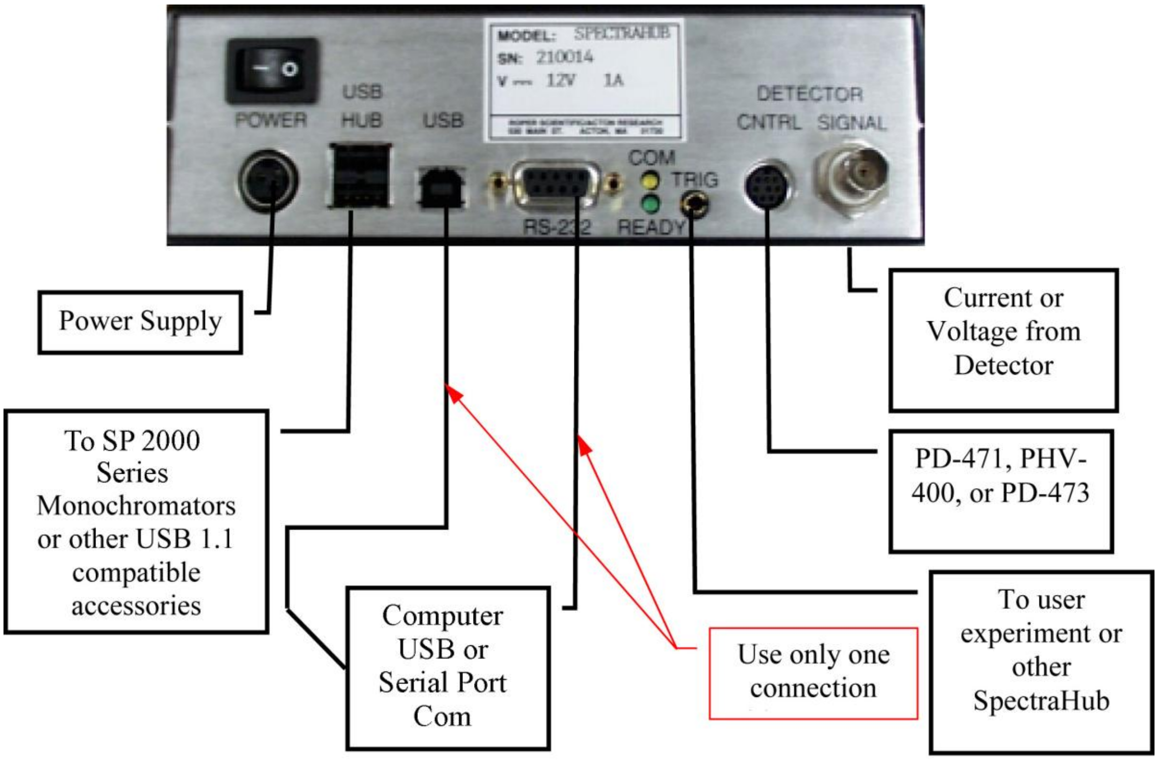
II. Installation and Setup

The SpectraHub is supplied with the following items:

SpectraHub Detector Interface Module
12 Volt DC power supply
AC line cord
Serial communications cable
USB communication cable
Operators Manual

Shown below is a diagram illustrating the connections of the SpectraHub with the Acton SP series monochromator and other Acton accessories such as the PD-471 PMT detector and the PD-473 Photon Counting detector plus the connection to the user's computer.

SpectraHub Rear Panel



If an experiment utilizes more than one Acton detector, multiple SpectraHubs can be triggered simultaneously by connecting the trigger lines together. Multiple SpectraHubs are controlled by connecting the SpectraHubs to the computer USB ports, computer RS-232 or any USB hub in the system. Note that the SpectraHub and all SP-2000 series SpectraPro monochromators have built-in USB hubs.

III. Specifications

Data Acquisition: Single 20-bit A/D converter

Inputs:

Current from a PMT, Silicon Diode or IR detector
Voltage from a preamplifier
Photon counting TTL pulses from an amplifier-discriminator such as the Acton model #PD-473-1 Photon Counting detector module

Sensitivity:

Current: +/-10nA to +/-1.0µA full scale
Voltage: +/-100mV to +/-10V full scale
(Note: Full scale readings are dependent on range setting and Integration time.)
Photon Counting: 20 nanosecond pulse pair resolution or up to 35 MHz continuous frequency input

Integration Time: 5 ms to 64 seconds

Control of PMT High Voltage:

Single D/outputs with 12-bit resolution providing the control voltage to the Acton PD-471 or PHV-400
Actual output voltage: 0 to +5VDC or 0 to +1.25 VDC

Communications:

USB 1.1 or
RS-232 baud rate 9600

Power:

12 Volts DC @ 750 amps from separate power supply provided with SpectraHub

Size: 7 in. (17.78 cm) wide x 2.3 in. (5.84 cm) high x 5.3 in. (13.46 cm) deep

IV. Operation

The SpectraHub typically is operated from a PC running Acton SpectraSense software version 4.3 or later and in conjunction with other Acton modules to comprise a complete spectroscopic system. Refer to the SpectraSense Operation Manual for a complete description of this system software. When beginning operation of the system, the modules of the systems must all be powered on. While this can occur in any order, the following turn-on sequence is recommended.

1. light source
2. monochromators
3. filter wheels and other accessories
4. SpectraHub
5. computer

The SpectraSense software provided with the SpectraHub will guide the operator through the normal spectroscopic data acquisition modes. For users who want to control their system through their own user-generated software, the ARC_Instrument.dll is provided on the SpectraSense install disk. The install disk contains a Software Developers Kit (SDK) with the dll and example code in Delphi, Visual Basic, C++ and Lab VIEW. For test purposes and special applications, it is possible to control the SpectraHub directly through its own communications port using the commands found in Section V. This is most easily tested by connecting the RS-232 port of the SpectraHub to the COM port of the computer. Use the Hyper term program found in Windows Applications and set to 9600 baud, 1 stop bit, 8 data bits, no parity and no flow control. Refer to the command set in Section V for additional details.

The data acquisition section of the SpectraHub reads the signals from the detector in the system. The SpectraHub measures current from photomultipliers and silicon diodes and voltage from preamplifier circuits. The range and polarity of detector currents and voltages is user-selectable from the SpectraSense software and from the SpectraHub command set. The ranges in the SpectraSense software are as follows: X1, X2, X5 and X50. The full scale readings on each of these ranges are determined by the polarity and integration time. Polarity can be set to unipolar for negative current and voltage input from 0 to full scale (PMTs and some silicon diodes have negative current outputs) or bipolar current and voltage input for negative full scale to positive full scale. The full scale raw count output for unipolar readings is 0 to 2^{20} or 1048576 and the full scale raw count output for bipolar readings is -2^{19} to 2^{19} or –524288 to 524288. Note that unipolar provides one additional bit of resolution on negative inputs; however, bipolar must be used for all positive readings. Examples of maximum current and voltage inputs are shown below:

Full Scale Values for Unipolar Current vs. Integration Time

Range	10ms	100ms	1000ms
X1	$-2x10^{-5}$A	$-2x10^{-6}$A	$-2x10^{-7}$A
X2	$-1x10^{-5}$A	$-1x10^{-6}$A	$-1x10^{-7}$A
X5	$-4x10^{-6}$A	$-4x10^{-7}$A	$-4x10^{-8}$A
X50	$-4\ x10^{-7}$A	$-4x10^{-8}$A	$-4x10^{-9}$A

(e.g., X1 range, 100 ms integration time, $-2x10^{-6}$A input = 1048576 raw counts)

Full Scale Values for Bipolar Current vs. Integration Time

Range	10ms	100ms	1000ms
X1	+1 to $-2x10^{-5}$A	+1 to $-2x10^{-6}$A	+1 to $-2x10^{-7}$A
X2	+/-$1x10^{-5}$A	+/-$1x10^{-6}$A	+/-$1x10^{-7}$A
X5	+/-$4x10^{-6}$A	+/-$4x10^{-7}$A	+/-$4x10^{-8}$A
X50	+/-$4x10^{-7}$A	+/-$4x10^{-8}$A	+/-$4x10^{-9}$A

(Note: X1 is limited to half the expected value for positive input or 262144 raw counts.)

Full Scale Values for Unipolar Voltage vs. Integration Time

Range	10ms	100ms	1000ms
X1	*	-5V	-0.5V
X2	*	-2.5V	-0.25V
X5	-10V	-1V	-0.1V
X50	-100mV	-100mV	-10.0mV

(Note: Input voltages greater than +/-10 Volts should not be used)

Full Scale Values for Bipolar Voltage vs. Integration Time

Range	10ms	100ms	1000ms
X1	*	+2.5 to -5.0V	+0.25V to -0.5V
X2	*	+/-2.5V	+/-0.25V
X5	+/-10V	+/-1V	+/-0.1V
X50	+/-1.0V	+/-100mV	+/-10.0mV

(Note: Input voltages greater than +/- 10Volts should not be used and positive voltages on X1 range are limited to half the expected value or 262144 raw counts.)

The SpectraHub also measures the counts from photon counting modules that include an internal amplifier/discriminator such as the Acton model PD-473 side window photon counting module. The photon counting input circuits in the SpectraHub will measure pulse pair resolutions to 20 ns or continuous frequency input to approximately 35 MHz. However, as the light intensity increases random distribution of pulses causes some pulses to overlap, affecting linearity. Generally, as the average frequency approaches 2 to 3 MHz, pulse overlap begins to occur. As the light intensity increases, the pulse overlap or “pulse pileup” can become severe enough to cause the output to go to zero. Photon counting is intended for low level light readings. Caution is advised as intensity increases since results can be misleading.

Each SpectraHub has associated with it a 12-bit D/A converter for controlling the PMT voltage in the Acton PMT detector modules such as the PD-471 or in the PMT high voltage supply, PHV-400. This voltage is controlled by setting the PMT high voltage value in the SpectraSense software or by the VOLTS command in the SpectraHub command set. The maximum value of the D/A voltage output is settable and stored in non-volatile memory in the SpectraHub. This voltage will be set to 1250 mV for the PD-471 or 5000 mV for the PHV-400 if one of these units is ordered with the SpectraHub. Each of these values will result in a maximum settable voltage of -1250 Volts at the PMT. If neither is ordered with the SpectraHub, the default value of 1250 is used. The maximum settable value for the D/A converter is 5000 mV.

There is one TTL trigger input and one TTL trigger output in the TRIG connector on the rear panel of the SpectraHub. Refer to Section VI of this manual for a diagram of the connector and description of each line. These inputs and outputs can be used to synchronize the user’s experiment with the actions of the SpectraHub. These lines are controllable through the SpectraSense software and by commands in the SpectraHub command set as described in Section VI. The power supply for the isolated side of the I/O is located in the SpectraHub and isolated from the remainder of the supplies. The inputs are TTL or switch contact compatible and the outputs are TTL with the capability of sinking 10 mA.

V. Command Set

I. Detector Readout Commands

? ENUM returns 1if the SpectraHub is set up as Channel 1 readout or if only one SpectraHub is ordered. If more than one SpectraHub is ordered, this returns the channel number assigned to this unit.

SET-ENUM   sets the channel number when more than one SpectraHub is used in a system (e.g., 2 SET-ENUM would set up a SpectraHub as a channel 2 readout).

READ takes 1 reading from the SpectraHub.

READS takes n readings and returns the read number and the value.

DELAY ms between readings (e.g., 100 DELAY inserts a 100 ms delay between each reading when using the READS command).

?DELAY returns the READS delay in ms.

TRIG-READ waits for a hardware trigger on the trigger input line and takes one reading

TRIG-READS   waits for a hardware trigger for each reading and returns the number of the read and the value.

VOLTAGE   sets the SpectraHub to read voltage at its input.

CURRENT   sets the SpectraHub to read current at its input.

PHOTON   sets input for selected channel to accept photon counting signals.

?READOUT returns the string "voltage", "current" or "photon" corresponding to the SpectraHub setting.

ITIME sets integration time for all channels in ms.

?ITIME returns the value for the integration time.

RANGE sets A/D converter gain for selected channel.

0 RANGE	corresponds to X1 range in SpectraSense
1 RANGE	corresponds to X2 range in SpectraSense
2 RANGE	corresponds to X5 range in SpectraSense
3 RANGE	corresponds to X50 range in SpectraSense

?RANGE returns A/D converter gain for selected channel. See RANGE for values.

UNI-POLAR sets the polarity for the A/D converter in the selected channel for negative unipolar readings.

BI-POLAR   sets the polarity for the A/D converter in the selected channel for bipolar or positive readings.

?POLARITY returns UNIPOLAR or BIPOLAR.

II. PMT High Voltage Commands

VOLTS sets the PMT voltage for the selected channel.

?VOLTS returns the PMT voltage for the selected channel.

HV-ON turns on the PMT supply and the PHOTON COUNTER supply.

HV-OFF   turns off the PMT and Photon counter supplies.

HV-REF sets the maximum value of the PMT control voltage in milliVolts. Use 1250 for PD-471 and 5000 for PHV-400.

?HV-REF returns the setting for max value of control voltage.

MAX-VOLTS sets the maximum value of the PMT high voltage output corresponding to the HV-REF setting. This value is 1250 for the PD-471, 1250 for the PHV-400 and 1800 for PHV-400-1800.

?MAX-VOLTS returns the value of MAX-VOLTS.

III. Shutter Commands

OPEN-SHUTTER opens shutter.

CLOSE-SHUTTER closes shutter.

?SHUTTER returns OPEN or CLOSED.

ENABLE-SHUTTER enables shutter operation.

DISABLE-SHUTTER disables shutter operation.

?SHUTTER-VALID returns "Shutter set up" or "Shutter not set up".

INIT-SHUTTER-LOW sets shutter output to TTL low for shutter open.

INIT-SHUTTER-HIGH sets shutter output to TTL high for shutter open.

?SHUTTER-HIGH returns the TTL open level setting.

?SHUTTER-INFO returns state of shutter setup state, open TTL state and open or closed.

IV. Readout System Commands

VER returns string representing software version.

MODEL returns string representing model # of readout system.

SERIAL returns string representing serial # of readout system.

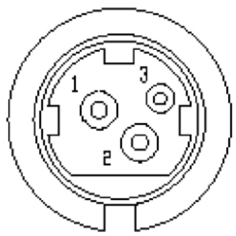
WHO returns Acton header for SpectraHub.

?EE-STATUS returns status of parameters in EEPROM.

VI. Connector Description

POWER CONNECTOR

Pin#	**Description**
1---------------	+12 VOLTS
2---------------	GND
3---------------	+12 VOLTS



RS-232 CONNECTOR

Pin#	**Description**
2--------------	RDX
3--------------	TDX
5--------------	GND

Note: Standard one-to-one cable may be used.

CONTROL CONNECTOR

Pin # **Description**

1------------- +5V Digital Switched

2------------- PMT Control Voltage

3------------- +15V

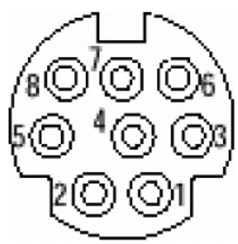
4------------- Analog Ground

5------------- Photon Counting In

6------------- Digital Ground

7------------- -15V

8------------- +5V Analog



TRIGGER CONNECTOR

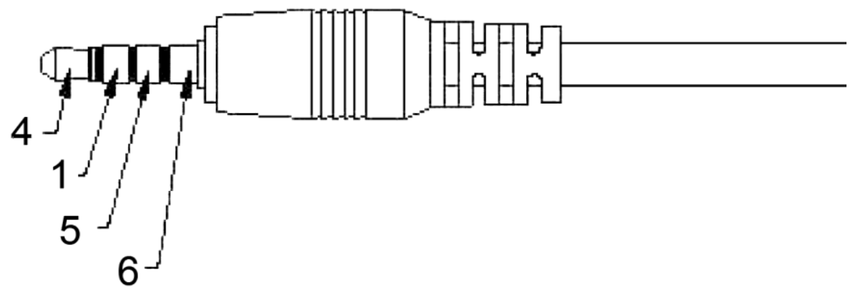
Pin # **Description**

Pin 1------------- Trigger in

Pin 4 ------------ Trigger out

Pin 5------------- Shutter TTL control – TTL level settable for open state

Pin 6 ------------- Ground

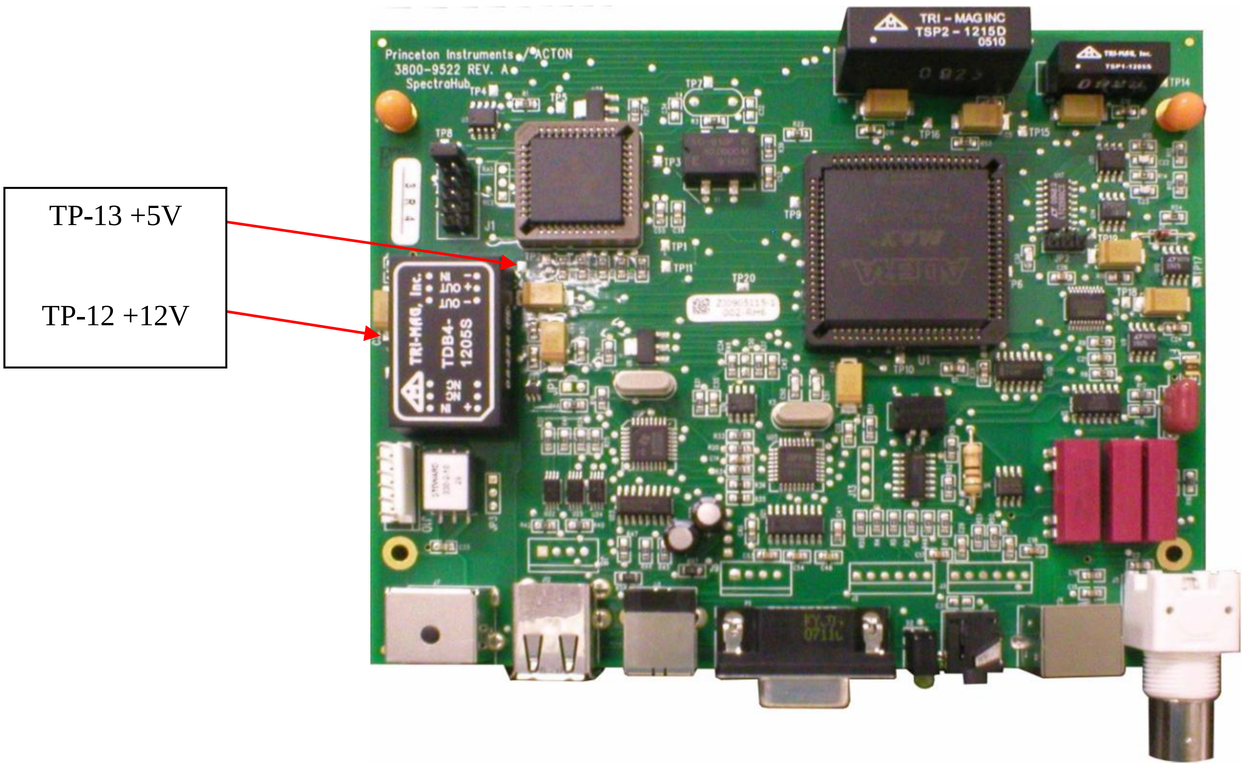


Mating cable and connector shown above are CUI Inc. part# CA354S

VII. Troubleshooting

The SpectraHub operates from an external +12 Volt DC supply provided with the SpectraHub at time of shipment. This +12 Volts DC is converted to +5VDC digital, +5VDC analog, and +/-15VDC analog voltages internally using board-mounted isolated DC-to-DC converters. The SpectraHub operates using an internal 32-bit microprocessor. Shown below are a list of possible symptoms, what to check and action to take. Refer to the photo showing the TP12 and TP13 test points.

	Symptom	Check
1.	Green power indicator LED does not come on when SpectraHub is switched on.	Make sure power supply is connected to an AC source and the power supply is connected to the SpectraHub. If connected properly, check the +12V and +5V at TP12 and TP13 on the SpectraHub pc board.
2.	The SpectraSense software cannot find the SpectraHub.	Make sure amber communications light on the rear panel of the SpectraHub flashes as the SpectraHub is switched on. Set the computer to terminal emulation mode at 9600 baud and test communications.
3.	SpectraSense software cannot find monochromators.	Be sure monochromator is on and fully initialized. Connect the computer in terminal emulation mode directly to the monochromator and test communications.


TP-13 +5V
TP-12 +12V
Princeton Instruments / ACTON
3800-9522 REV. A
SpectraHub
TRI - MAG INC
TSP2 - 1215D
TRI-MAG, Inc.
TDB4-
1205S